

Muhammad Ahmad

Full Stack | AI / ML Engineer

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PROFESSIONAL SUMMARY

Full Stack AI Engineer with over 3 years of professional experience designing and deploying intelligent systems across computer vision, machine learning, and large language model (LLM) architectures. Proven track-record building production-grade RAG pipelines, agentic reasoning workflows (Chain-of-Thought, ReAct), and real-time voice-enabled AI services using Python and FastAPI. Hands-on expertise with deep learning model training (CNNs, YOLO), cloud deployment on AWS, and vector database integration. Published researcher in AI-based cybersecurity.

SKILLS

Programming Languages: Python, JavaScript, SQL

Frameworks & Libraries: FastAPI, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, PyMuPDF, Electron.js, Bootstrap, Ruby on Rails

AI & Machine Learning: Large Language Models (LLMs), RAG Pipelines, Chain-of-Thought (CoT), ReAct, CNNs, YOLO (v8–v11), Computer Vision, NLP, KNN, Naïve Bayes, Regression, Decision Tree

Databases & Vector Stores: PostgreSQL, MySQL, ChromaDB, FAISS

Cloud & DevOps: AWS (EC2, ECR, ECS, S3, Cognito), Azure OpenAI, Docker

APIs & Tools: GPT-4 Realtime API, Whisper ASR, Groq LLM API, IPC, Git

WORK EXPERIENCE

Careonix

January 2026 – Present

Full Stack AI Engineer

Lahore, Pakistan

- Design and deploy agentic AI systems leveraging LLM reasoning, tool-calling, and retrieval-augmented generation (RAG) to automate complex clinical workflows.
- Build real-time, voice-enabled AI pipelines using Azure OpenAI (GPT-4 Realtime, Whisper ASR) and FastAPI streaming APIs, reducing average query response latency by over 40%.
- Develop and maintain cloud-native AI services on AWS (ECS, EC2, S3) with Docker containerization and Cognito-based authentication, achieving 99.9% uptime in production.
- Integrate PostgreSQL and vector databases to power scalable semantic search and context retrieval across large document corpora.

Cyberautix Technologies

March 2023 – December 2025

Software Engineer

Lahore, Pakistan

- Architected and shipped AI-powered desktop applications using Electron.js, bridging FastAPI Python inference services via IPC for secure, low-latency decision-making and workflow automation.
- Built end-to-end computer vision and document-processing pipelines using TensorFlow, YOLO (v8–v11), OpenCV, and PyMuPDF — covering dataset curation, model training, evaluation, and productionization across 5+ client projects.
- Optimized image-processing pipelines with NumPy vectorization and OpenCV, improving model inference speed by approximately 30% and increasing detection accuracy by 15%.
- Delivered 3+ production-ready desktop AI applications, each integrating custom-trained models via REST APIs, directly enhancing client operational efficiency.

Devsinc

December 2022 – February 2023

Software Engineer Intern

Lahore, Pakistan

- Developed and maintained Ruby on Rails web application features, contributing to a codebase serving thousands of active users.
- Implemented responsive UI components using Bootstrap, improving cross-device compatibility across the product.
- Participated in daily code reviews and Agile sprint ceremonies, adopting professional engineering practices under senior mentor guidance.

KEY PROJECTS

OnixMend | FastAPI, PostgreSQL, pgvector, RAG, LLMs, AWS

- Built a production multi tenant AI knowledge platform enabling organizations to securely upload internal documents and interact with an AI assistant powered by Retrieval Augmented Generation and vector search.

OnixAsk | FastAPI, PostgreSQL, pgvector, RAG, LLMs, Docker

- Built a multi-tenant AI knowledge platform enabling organizations to deploy branded, website-embeddable chatbots trained on their own documents using RAG-powered semantic search..

AI Voice Receptionist for Dental Clinics | FastAPI, Azure OpenAI, Whisper ASR, AWS, RAG

- Developed an AI powered voice receptionist capable of understanding caller intent, managing appointment workflows, answering clinic specific questions using Retrieval Augmented Generation, and automating patient communication.

InsightLens | Python, FastAPI, ChromaDB, RAG, LLMs

- Designed and developed an AI-powered insight-generation platform integrating RAG pipelines with LLM reasoning, enabling users to query large document collections with high accuracy.

CricketSense | Python, FAISS, Groq LLM API, CoT, ReAct

- Built a semantic cricket reasoning assistant using FAISS vector retrieval and Chain-of-Thought (CoT) prompting with ReAct reasoning, achieving contextually accurate responses on complex sports queries.

Musculoskeletal Radiograph Abnormality Detection System | TensorFlow, CNNs

- Developed an automatic x-ray classifier by custom models training and integrating multiple CNN models into a production-ready desktop application for automated medical image classification and abnormality detection.

Published Research | Springer - "Classification of Cloud Attacks Using Deep Learning"

- Applied deep learning models to classify cloud network traffic for cybersecurity threat detection; research accepted and published in Springer academic journal.

EDUCATION

University of Central Punjab

2019 – 2023

Bachelor of Science in Computer Science (BSCS)

Lahore, Pakistan

CGPA: 3.76 / 4.0

Superior College, Multan Road Campus

2017 – 2019

Intermediate in Computer Science (ICS)

Lahore, Pakistan

CERTIFICATIONS

- Software Engineer-HackerRank
- RAG Course for Beginners – Simplilearn
- Artificial Intelligence Course – Samsung Innovation Campus
- Ruby on Rails Bootcamp – Devsinc
- Programming Foundations with JavaScript, HTML, and CSS – Coursera